

Case Project: Forensic Investigation

[REDACTED]

[REDACTED]

Somsouk Thepphavong [REDACTED]

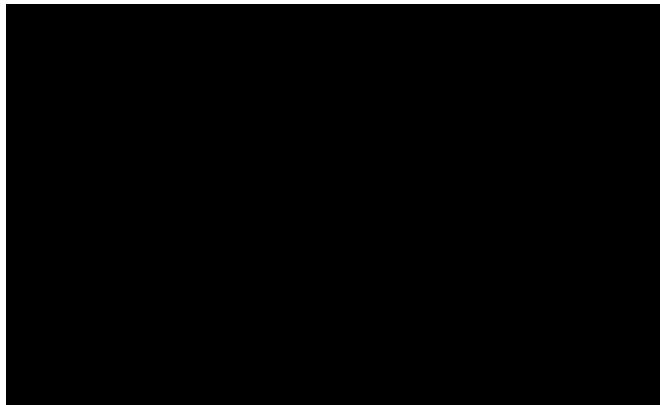


Table of Contents

1. Introduction	1
2. Investigation Process	2
3. File Images Integrity	3
4. File Image Overview.....	4
5. Findings	5
5.1. Computer & Account Information	5
5.2. Time Zone Setting	6
5.3. Application.....	7
5.3.1. Executable Program	7
5.3.2. Email.....	8
5.3.3. Browser	8
5.3.4. External Storage	9
5.4. Deleted and Traversed Files	10
5.4.1. Recycling Bin	10
5.4.2. Deleted Files	10
6. Evidence Analysis	11
6.1. Email Communication.....	11
6.2. External Storage Examination	11
6.3. Web Search History.....	13
7. Limitation.....	15
8. Timeline.....	16
9. Conclusion	17
References	18
Appendix A.....	19
Appendix B	20

1. Introduction

This report presents a comprehensive digital forensic investigation into suspected corporate data theft involving Mr. Iaman Informant, an employee suspected of unauthorised disclosure of confidential company information. The investigation was initiated following the confiscation of suspicious removable media devices at a security checkpoint, prompting a thorough examination of the suspect's office computer and associated storage devices.

The primary objective of this investigation is to determine whether proprietary company data, specifically designated "Seed Files," was unlawfully extracted and transferred to external parties. Through systematic analysis of multiple evidence sources, including the suspect's personal computer, two USB drives (Removable Media #1 and #2), and one CD-ROM (Removable Media #3), this investigation aims to establish a clear timeline of events, identify methods of data exfiltration, and uncover any anti-forensic techniques employed to conceal criminal activities.

Utilising industry-standard forensic tools including FTK Imager, Autopsy, Eric Zimmerman's Tools suite, and various command-line utilities, this investigation follows established digital forensic methodologies to ensure the integrity and admissibility of all collected evidence. All timestamps referenced throughout this report are presented in Coordinated Universal Time (UTC) unless otherwise specified, maintaining consistency and precision in analysis.

Particular attention is paid to artefacts indicating file deletion patterns, external device connectivity, cloud storage synchronisation, and evidence of deliberate evidence tampering. The findings presented herein provide a substantial basis for determining whether corporate security policies were violated and confidential information was compromised.

2. Investigation Process

In this comprehensive analysis, we will utilise suitable forensic tools (Yaacoub et al., 2021, p. 18).

Here are the listed tools being used in this forensic investigation

- FTK Imager
- Autopsy
- Eric Zimmerman's Tools
- Command Prompt
- PowerShell

3. File Images Integrity

After acquiring the file images needed for forensic investigation, we will check the file's hash value and compare it with the official's hash value. This method will ensure that the file images are not altered and that the content inside those files is not tampered with.

Data Leakage Case

SHA1 for Download Files

File Name	SHA1
cfreds_2015_data_leakage_pc.7z.001	F07632FAA66A47088DE0780B45CC568E4BF650B
cfreds_2015_data_leakage_pc.7z.002	50EE46ABF6FA833268E5AE199A13854CCF42689B
cfreds_2015_data_leakage_pc.7z.003	1687686F819092E05047F195F102D8FABC38ED66
cfreds_2015_data_leakage_pc.E01	72432916933F5A309A8C456840C9601D1F8D2A4F
cfreds_2015_data_leakage_pc.E02	0CAF4261ED8432A8B38AA019B1B28DF96F79130
cfreds_2015_data_leakage_pc.E03	BE836C891736C4C0C2253C6803399BF0F2A599BA
cfreds_2015_data_leakage_pc.E04	91598FFD56097495F73F8BF967875EB288B1E3DE
cfreds_2015_data_leakage_rm#1.E01	FFD0F3CBA3DFE3291F786B845A06A8AA56C1CD8C
cfreds_2015_data_leakage_rm#2.7z	DDFE97AA3D8D0B33CC6092123090A8154945F38E

cfreds_2015_data_leakage_pc.E01	72432916933F5A309A8C456840C9601D1F8D2A4F
cfreds_2015_data_leakage_pc.E02	0CAF4261ED8432A8B38AA019B1B28DF96F79130
cfreds_2015_data_leakage_pc.E03	BE836C891736C4C0C2253C6803399BF0F2A599BA
cfreds_2015_data_leakage_pc.E04	91598FFD56097495F73F8BF967875EB288B1E3DE
cfreds_2015_data_leakage_rm#1.E01	FFD0F3CBA3DFE3291F786B845A06A8AA56C1CD8C
cfreds_2015_data_leakage_rm#2.7z	DDFE97AA3D8D0B33CC6092123090A8154945F38E
cfreds_2015_data_leakage_rm#2.E01	2228554CD6FD03C85B8B0E0A0CD7F21A2364DC99

cfreds_2015_data_leakage_rm#1.E01	FFD0F3CBA3DFE3291F786B845A06A8AA56C1CD8C
cfreds_2015_data_leakage_rm#2.7z	DDFE97AA3D8D0B33CC6092123090A8154945F38E
cfreds_2015_data_leakage_rm#2.E01	2228554CD6FD03C85B8B0E0A0CD7F21A2364DC99
cfreds_2015_data_leakage_rm#3_type1.7z	13EEDDC40AD493022B2D7E0A6674DEF603F6247
cfreds_2015_data_leakage_rm#3_type2.7z	AE26235F6F85EDDFB670DD060EF109EDA91E88F

cfreds_2015_data_leakage_rm#2.E01	2228554CD6FD03C85B8B0E0A0CD7F21A2364DC99
cfreds_2015_data_leakage_rm#3_type1.7z	13EEDDC40AD493022B2D7E0A6674DEF603F6247
cfreds_2015_data_leakage_rm#3_type2.7z	AE26235F6F85EDDFB670DD060EF109EDA91E88F
cfreds_2015_data_leakage_rm#3_type3.E01	75C106FDB2FD2F8068190E951589FF1F9860257E
cfreds_2015_data_leakage_seed_files.7z	E79C8C245F990EA8E79D9A099B0AA7A1E78EC0DE

cfreds_2015_data_leakage_rm#3_type2.7z	AE26235F6F85EDDFB670DD060EF109EDA91E88F
cfreds_2015_data_leakage_rm#3_type3.E01	75C106FDB2FD2F8068190E951589FF1F9860257E
cfreds_2015_data_leakage_seed_files.7z	E79C8C245F990EA8E79D9A099B0AA7A1E78EC0DE

```
PS C:\Users\somso\OneDrive\Desktop\File Image\Personal PC> Get-ChildItem "C:\Users\somso\OneDrive\Desktop\File Image\Personal PC" | Get-FileHash -Algorithm SHA1
```

Algorithm	Path	Hash
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Personal PC	72432916933F5A309A8C456840C9601D1F8D2A4F
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Personal PC	0CAF4261ED8432A8B38AA019B1B28DF96F79130
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Personal PC	BE836C891736C4C0C2253C6803399BF0F2A599BA
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Personal PC	91598FFD56097495F73F8BF967875EB288B1E3DE

```
PS C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 1 (No Need to Analyze)\cfreds_2015_data_leakage_rm#2.E01> Get-FileHash -Algorithm SHA1
```

Algorithm	Path	Hash
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 1 (No Need to Analyze)\cfreds_2015_data_leakage_rm#2.E01	2228554CD6FD03C85B8B0E0A0CD7F21A2364DC99

```
PS C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 2 (USB 2.0)\cfreds_2015_data_leakage_rm#2.E01> Get-FileHash -Algorithm SHA1
```

Algorithm	Path	Hash
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 2 (USB 2.0)\cfreds_2015_data_leakage_rm#2.E01	2228554CD6FD03C85B8B0E0A0CD7F21A2364DC99

```
PS C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 3 (CD-ROM)\cfreds_2015_data_leakage_rm#3_type1.7z> Get-FileHash -Algorithm SHA1
```

Algorithm	Path	Hash
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 3 (CD-ROM)\cfreds_2015_data_leakage_rm#3_type1.7z	13EEDDC40AD493022B2D7E0A6674DEF603F6247
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 3 (CD-ROM)\cfreds_2015_data_leakage_rm#3_type2.7z	C11F21DE9C8AE60877F89B82ACFA1F92D9C85F8F
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 3 (CD-ROM)\cfreds_2015_data_leakage_rm#3_type3.E01	2581F0C08A992E531EBF0D4AA1143C6B3730ED02
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Removeable Media 3 (CD-ROM)\cfreds_2015_data_leakage_seed_files.7z	99C7ACE8ECD0D7E506CEBBE41623299E05846A60

```
PS C:\Users\somso\OneDrive\Desktop\File Image\Seed File> Get-FileHash -Algorithm SHA1
```

Algorithm	Path	Hash
SHA1	C:\Users\somso\OneDrive\Desktop\File Image\Seed File	E79C8C245F990EA8E79D9A099B0AA7A1E78EC0DE

By using the PowerShell command, we generate a SHA1 hash value (Boyanov, 2024, pp. 122-125).

Get-ChildItem "[Directory of the file]" | Get-FileHash -Algorithm SHA1

This will generate a hash of the content in that directory. Then we compare the official hash value. This way, we can be sure that our image files are identical.

4. File Image Overview

We have gathered all file images for each media storage device:

- Personal Computer
- Removable Media # 1 (USB Drive)
- Removable Media # 2 (USB Drive)
- Removable Media # 3 (CD-ROM)

Personal Computer

- This is the suspect's office computer, used during office hours. It holds a variety of information regarding his illegal file transfer. Including email communication, Google Drive artifacts, Web Browser history, deleted images and etc.

Removable Media # 1

- This is the company's authorised peripheral and could store any company's valuable information. Therefore, any information stored in this media storage will not raise any suspicion.
- It stored the "Seed File" of the company's confidential files. Assuming these "Seed Files" are found elsewhere, confirmation of data leakage.

Removable Media # 2

- This is the primary scope of this digital forensic investigation.
- The suspicious unauthorised USB drive was confiscated at the security checkpoint.

Removable Media # 3

- An external CD-ROM was confiscated at the security checkpoint.

5. Findings

5.1. Computer & Account Information

- Windows 7 Ultimate (SP1)
- Registered Owner – Iaman Informant
- PC Name – Informant-PC
- Processor Architecture AMD64
- Install Date – 1427034866 in Unix epoch
 - o Convert to UTC – 2015-03-22 07:14:26.

Registry Explorer v2.1.0

File Tools Options Bookmarks (32/3) View Help

Registry Hives (1) Available bookmarks (32/3)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
Router	0	0	2009-07-14 04:49:00
Rpc	2	3	2015-03-25 13:29:46
SchedulingAgent	4	0	2009-07-14 05:08:49
Schema Library	0	1	2015-03-22 15:03:49
Security Center	1	2	2015-03-25 15:31:00
Sensors	1	0	2010-11-21 07:16:39
Shared	0	3	2015-03-22 15:04:16
Shared Tools	1	6	2015-03-22 15:03:49
Shared Tools Location	1	0	2015-03-22 15:18:29
SideShow	1	2	2010-11-21 07:16:39
Software	0	1	2009-07-14 04:49:00
Speech	0	0	2010-11-21 07:06:35
SQMClient	1	3	2015-03-25 10:15:44
Sync Framework	0	1	2009-07-14 04:49:00
Synopsis	2	0	2015-03-25 10:15:21
SystemCertificates	0	10	2015-03-23 15:00:47
TabletService	0	4	2009-07-14 04:49:00
TabletTip	2	2	2010-11-21 07:17:14
TabletTip	0	1	2009-07-14 04:53:18
Terminal Server Client	0	3	2009-07-14 04:49:00
TermServ Licensing	0	0	2009-07-14 04:49:00
TSP Shared	0	1	2009-07-14 04:49:00
TPM	0	0	2010-11-21 07:16:39
TPM	0	1	2009-07-14 04:49:00
Tracing	1	10	2015-03-22 15:54:00
Transaction Server	0	1	2009-07-14 04:49:00
TV System Services	0	1	2009-07-14 04:49:00
UDRM	1	0	2009-07-14 04:49:00
UPnP Device Host	0	3	2015-03-22 15:18:33
VIA	1	0	2015-03-22 15:03:49
Virtual Machine	0	0	2010-11-21 07:16:29
WAB	0	6	2009-07-14 04:49:00
WDDM	3	7	2010-11-21 03:33:36
WDDM	0	0	2009-07-14 04:49:00
Windows	0	10	2015-03-22 15:00:57
Windows Defender	4	10	2015-03-23 10:17:57
Windows Desktop Search	1	0	2010-11-21 03:33:36
Windows Mail	5	4	2010-11-21 03:33:36
Windows Media Device Manager	1	3	2009-07-14 04:49:00
Windows Media Foundation	0	5	2009-07-14 04:49:00
Windows Media Player NSS	0	1	2009-07-14 04:49:00
Windows Messaging Subsystem	6	1	2015-03-22 15:01:37
Windows NT	0	1	2009-07-14 04:53:25
CurrentVersion	21	20	2015-03-22 15:23:53
Windows Defender	0	1	2010-11-21 04:49:00

Values

Drop a column header here to group by that column

Value Name	Value Type	Data	Value Size	Is D.	Data Record Reallocated
InstallDate	RegDword	1427034866			
RegisteredOrganization	RegSz				
DigitalProductId	RegBinary	A4-00-00-00-00-00-00-00-30-30-34-32-36-20-32-39-32-3D-30-3...			
CurrentBuild	RegSz	7601	00-00		
EditionID	RegSz	Ultimate	00-00		
CurrentBuildNumber	RegSz	7601	00-00		
BuildLab	RegSz	7601.win7p1_gdr.130828-1532	00-00		
BuildID	RegSz	cefa176-8b62-4aee-w8ff-5c9dc9ade1d	00-00		
CSDBuildNumber	RegSz	1130	00-00		
DigitalProductId4	RegBinary	P8-04-00-00-04-00-00-00-30-30-30-34-00-32-00-36-00-20-00...	00-00-00-00		
SoftwareType	RegSz	System	00-00-00-00		
SystemRoot	RegSz	C:\Windows	00-00-00-00		
InstallerType	RegSz	Client	00-00-00-00		
ProductName	RegSz	Windows 7 Ultimate	00-00-00-00		
PathName	RegSz	C:\Windows	00-00-00-00		
BuildLabEx	RegSz	7601.18247.amd64fre.win7p1_gdr.130828-1532	00-00-00-00		
CurrentVersion	RegSz	6.1	38-0C-44-00		
CSDVersion	RegSz	Service Pack 1	39-00-30-00		
CurrentType	RegSz	Multiprocessor Free	65-00-64-00		
RegisteredOwner	RegSz	Iaman Informant	65-00-72-00		
ProductId	RegSz	00406-292-0000007-65282	69-63-72-4F		

Type viewer Binary viewer

Value name InstallDate

Value type RegDword

Value 1427034866

On the Registry Explorer tool, we can inspect the registry hive

C:\Windows\System32\config\SOFTWARE and navigate to the key Microsoft\Windows NT\

CurrentVersion.

5.2. Time Zone Setting

The screenshot displays the Windows Registry Editor. The left pane shows the tree structure expanded to **Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\TimeZones**. The right pane shows the **TimeZones** value, which is a string containing the name of the selected time zone: **Eastern Standard Time**.

Value Name	Value Data	Value Data Type
TimeZones	Eastern Standard Time	String

The time zone of the image is set in the Eastern Standard Time (UTC -5).

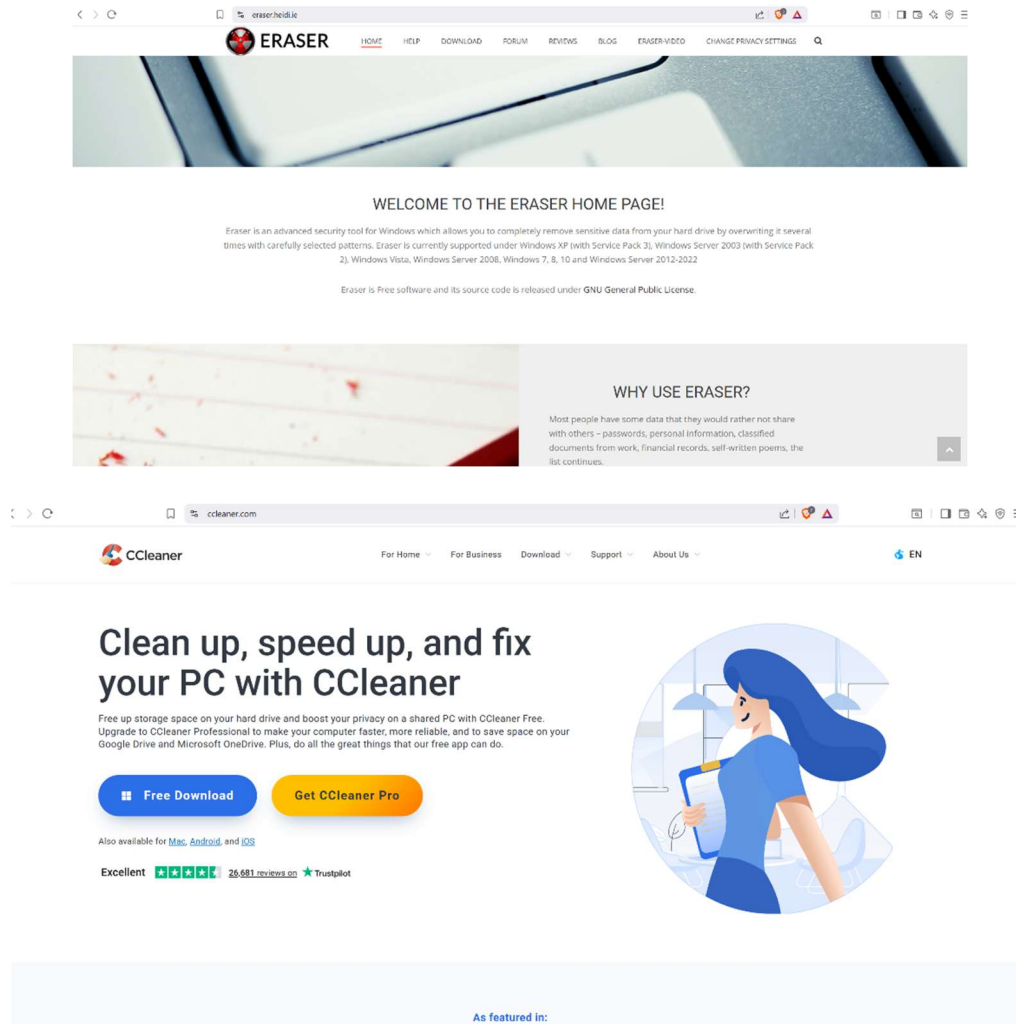
By analyzing the **SYSTEM registry file** (the file corresponding to the HKLM\SYSTEM hive), we identified the time zone. Using Registry Explorer.

5.3. Application

5.3.1. Executable Program

We found two executable files installed in the system, which could be responsible for large file deletion.

- Ccleaner
- Eraser



This is a powerful program used for file deletion, Eraser, that allows users to completely remove sensitive data from the hard drive by overwriting it several times with carefully selected patterns (Jones & Afrifa, 2020, p. 9). CCleaner helps clean up the files for optimisation and performance, and it can perform complete file deletion.

5.3.2. Email

- Outlook is used as an email communication platform based on the header

-----HEADERS-----

Received: from BN1PR09MB106.namprd09.prod.outlook.com (10.255.199.152) by
BL2PR09MB099.namprd09.prod.outlook.com (10.255.230.152) with Microsoft SMTP
Server (TLS) id 15.1.106.15 via Mailbox Transport; Mon, 23 Mar 2015 17:29:28
+0000

Received: from BN1PR09MB0156.namprd09.prod.outlook.com (10.255.192.146) by
BN1PR09MB106.namprd09.prod.outlook.com (10.255.199.152) with Microsoft SMTP
Server (TLS) id 15.1.99.14; Mon, 23 Mar 2015 17:29:27 +0000

Received: from BN1PR09MB0156.namprd09.prod.outlook.com ([10.255.192.146]) by

- We found two email addresses for both suspects
 - o Mr. Iaman Informant – iaman-informant@nist.gov
 - o Mr. Spy Conspirator – spy.conspirator@nist.gov

5.3.3. Browser

- A Windows default browser, Internet Explorer.
- Mr. Iaman Informant accessed the company's Secured Network Drive, IP address of 10.11.11.128, where "Seed Files" were stored.

Name	Location	Modified Time	Change Time	Access Time	Created Time
\$MFT	/img_cfred...	2015-03-25 11...	2015-03-25 ...	2015-03-25 ...	2015-03-25 11:08:36 UTC
\$UsnJml:\$J	/img_cfred...	2015-03-25 10...	2015-03-25 ...	2015-03-25 ...	2015-03-25 10:15:46 UTC
(secret_project)_pricing_decision.xlsx.LNK	/img_cfred...	2015-03-23 20...	2015-03-23 ...	2015-03-23 ...	2015-03-23 20:26:53 UTC
(secret_project)_pricing_decision.xlsx.lnk	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
(secret_project)_pricing_decision.xlsx.lnk	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
(secret_project)_pricing_decision.xlsx.lnk	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
(secret_project)_pricing_decision.xlsx.lnk	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
(secret_project)_pricing_decision.xlsx.lnk	/img_cfred...	2015-03-23 20...	2015-03-23 ...	2015-03-23 ...	2015-03-23 20:26:53 UTC
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00
0	/img_cfred...	0000-00-00 00...	0000-00-00 ...	0000-00-00 ...	0000-00-00 00:00:00

Hex

Text

Application

File Metadata

OS Account

Data Artifacts

Analysis Results

Context

Annotations

Other Occurrences

Strings

Extracted Text

Translation

Page: 1 of 1 Page

Matches on page: 1 of 5 Match

100%

Reset

(secret_project)_pricing_decision.xlsx.lnk \\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_project)_pricing_decision.xlsxA\\10
1SPS0
10.11.11.128
1SPS:
1SPSSC
\\10.11.11.128\secured_driveMicrosoft NetworkCompany's Secured Network Drive

5.3.4. External Storage

We were able to confirm that these devices were connected to the suspect PC.

Source Name	S	C	O	△ Date/Time	Device Make	Device Model	Device ID
SYSTEM			3	2015-03-24 13:38:00 UTC	SanDisk Corp.	Cruzer Fit	4C530012450531101593
SYSTEM			3	2015-03-24 13:38:00 UTC	SanDisk Corp.	Cruzer Fit	4C530012450531101593
SYSTEM			3	2015-03-24 19:38:09 UTC	SanDisk Corp.	Cruzer Fit	4C530012550531106501
SYSTEM			3	2015-03-24 19:38:09 UTC	SanDisk Corp.	Cruzer Fit	4C530012550531106501

- RM#1 - 4C530012450531101593
- RM#2 - 4C530012550531106501

When a CD disk is inserted into the PC, it creates a shortcut file named “CD Drive.lnk”. Indicating that a CD was running in the machine.

△ Source Name	S	C	O	Path	Date Accessed	Data Source
(secret_project)_pricing_decision.xlsx.LNK				\\10.11.11.128\SECURED_DRIVE\Secret Project Data\...	2015-03-23 20:26:53 UTC	cfreds_2015_data_leakage_pc.E01
(secret_project)_pricing_decision.xlsx.lnk				\\10.11.11.128\SECURED_DRIVE\Secret Project Data\...	0000-00-00 00:00:00	cfreds_2015_data_leakage_pc.E01
.lnk				E:\	0000-00-00 00:00:00	cfreds_2015_data_leakage_pc.E01
.lnk				D:\	0000-00-00 00:00:00	cfreds_2015_data_leakage_pc.E01
CD Drive (2).lnk				D:\	2015-03-24 21:01:11 UTC	cfreds_2015_data_leakage_pc.E01
CD Drive.lnk				D:\	2015-03-24 20:47:22 UTC	cfreds_2015_data_leakage_pc.E01
Desktop.LNK				C:\Users\informant\Desktop	2015-03-24 18:48:40 UTC	cfreds_2015_data_leakage_pc.E01

snapshot.db				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-23 20:02:51 UTC	2015-03-23 20:02:51 UTC
snapshot.db-shm				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:21:34 UTC
snapshot.db-wal				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:21:34 UTC
sync_config.db				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-23 20:02:51 UTC	2015-03-23 20:02:51 UTC
sync_config.db-shm				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:21:34 UTC
sync_config.db-wal				2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:22:48 UTC	2015-03-25 15:21:34 UTC
sync_log.log			1	2015-03-25 15:23:00 UTC	2015-03-25 15:23:00 UTC	2015-03-23 20:02:51 UTC	2015-03-23 20:02:51 UTC
sync_log.log				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

In Autopsy, navigate to `User\informant\AppData\Local\Google\Drive\user_default`. We found `sync_config.db` and `snapshot.db`. These files hold file details for files stored in the Google Drive account. Indicating that a file was uploaded to Google Drive (Quick & Choo, 2014).

```

\\?C:\Users\informant\Google Drive\happy_holiday.jpg
\\?C:\Users\informant\Google Drive\do_u_wanna_build_a_snow_man.mp3
w \\?C:\Users\informant\Google Drive\happy_holiday.jpgG
\\?C:\Users\informant\Google Drive\do_u_wanna_build_a_snow_man.mp3

```

These were files uploaded to Google Drive.

5.4. Deleted and Traversed Files

5.4.1. Recycling Bin

Recycle Bin								
Table Thumbnail Summary								
This is a DataResult window								
Source Name	S	C	O	Path	Time Deleted	Username	Data Source	
\$RKXD1U3.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RI3FM2A.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RIQGWT.ini				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$R508CBB.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RJEMT64.exe				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$R8YP3XK.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RU3FKWI.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RX538VH.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RFVCH5V.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	
\$RDOI3HE.jpg				C:\Users\informant\AppData\Local\Microsoft\Window...	2015-03-24 20:11:42 UTC		cfreds_2015_data_leakage_pc.E01	

We found 10 items inside the recycling bin. These items do not contain valuable information for this investigation.

5.4.2. Deleted Files

A total of 8436 files were deleted inside the machine.

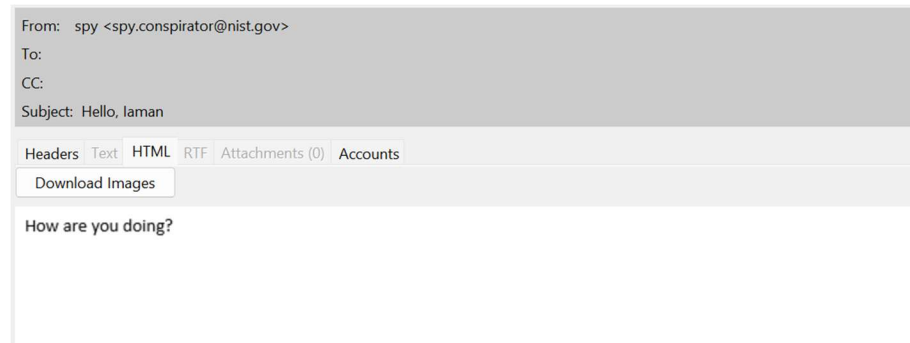
Listing Keyword search 1 - cleaner x											
All											
Table Thumbnail Summary											
Save Table as CSV											
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags	
7267.tmp~Rfb7242.TMP				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
7268.tmp				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
Session Storage				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
GreenBubbles.jpg				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
OrangeCircles.jpg				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
WMSDKNS.XML.bak				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
~usic.tmp				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
prf952C.tmp				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
prf952D.tmp				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
f_000010				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
f_000011				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
f_000012				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		
f_00001e				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		

This raises a question about the reason for deleting a large number of files.

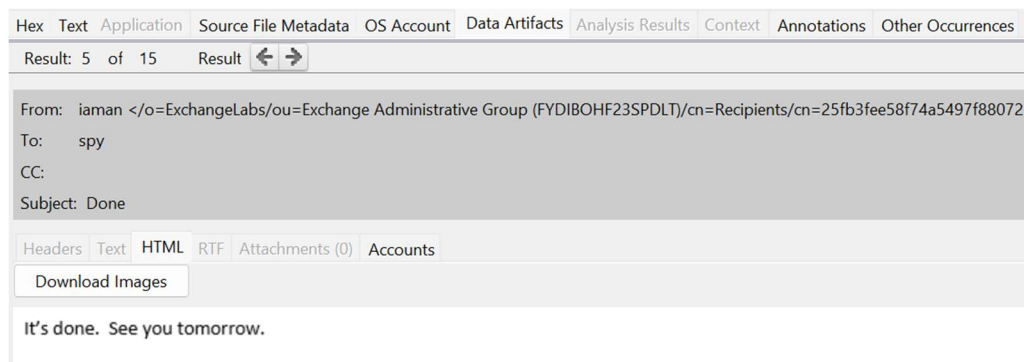
6. Evidence Analysis

6.1. Email Communication

On 23rd of March 2015, the first email was received from Mr. Conspirator, who initially sent an email greeting to Mr. Iaman offering high payment for the company's data leakage.



The conversation lasted until the 25th of March 2015. The content of the whole conversation was vague and often one-sided, indicating that more data was needed.



See **Appendix A** for the full conversation.

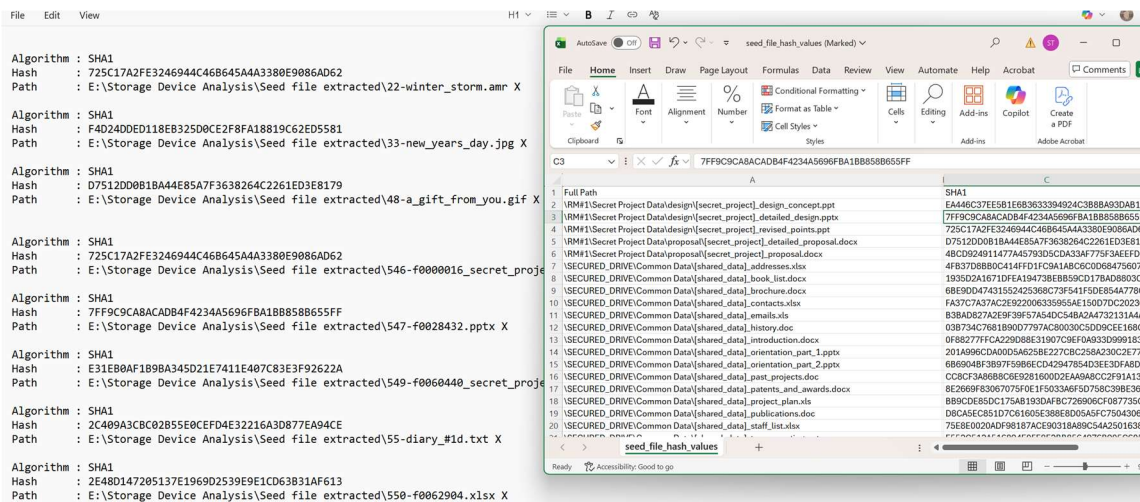
6.2. External Storage Examination

We narrowed our focus to RM#2 and RM#3, since RM#1 is an authorised USB drive, allowed to be carried and used during off-hours. Inside the RM#2, we found what we believed to be a “Seed Files”, knowing the company's secret files, used to claim those who possess these files are found guilty.

Inside the seed file, there are 39 files in total, each with its own hash value to confirm its integrity and prevent any alteration of its content. 22 of them are labelled as “secret project”.

18 out of 22, we have found in RM#2 are “secret project” labelled. 17 out of 22 seed files were retrieved and verified with the hash value. Using PowerShell command

Get-ChildItem [File Directory] | Get-FileHash -Algorithm SHA1, compared to the official hash table.



Confirming there was an actual file transition. See **Appendix B** for further explanation.

For RM#3 (CD-ROM), LECmd helps parse the .lnk file. We see that the CD Drive.lnk file was created, indicating that a CD was inserted into the system (Joshi et al., 2025, pp. 378-380).

```
Command line: -f .\CD Drive.lnk
Warning: Administrator privileges not found!

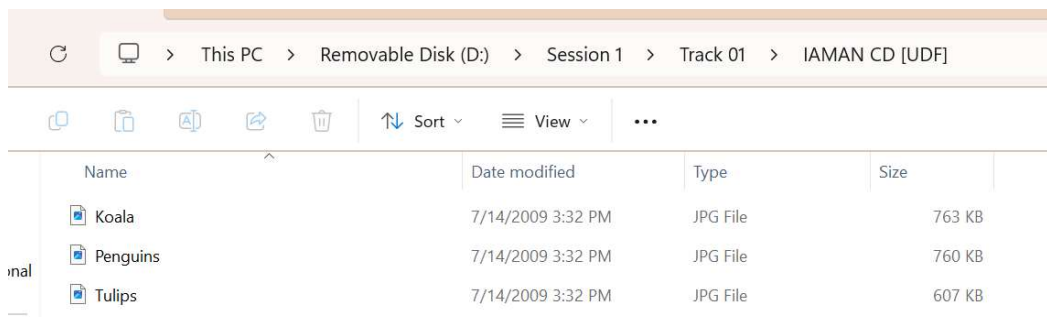
Processing E:\Storage Device Analysis\Personal PC\CD Drive.lnk

Source file: E:\Storage Device Analysis\Personal PC\CD Drive.lnk
Source created: 2025-11-01 05:40:33
Source modified: 2025-11-01 05:40:33
Source accessed: 2025-11-01 05:42:55

--- Header ---
Target created: 2015-03-24 20:44:53
Target modified: 2015-03-24 20:45:23
Target accessed: 2015-03-24 20:45:23

File size (bytes): 192
Flags: HasTargetIdList, HasLinkInfo, IsUnicode, DisableKnownFolderTracking
File attributes: FileAttributeReadOnly, FileAttributeDirectory
Icon index: 0
Show window: SwNormal (Activates and displays the window. The window is rest
the window is minimized or maximized.)
```

Upon inspecting Removable Media #3 (CD-ROM), it is in ISO format. FTK imager helped mount the .ISO file format, granting us Read-only access to the storage.



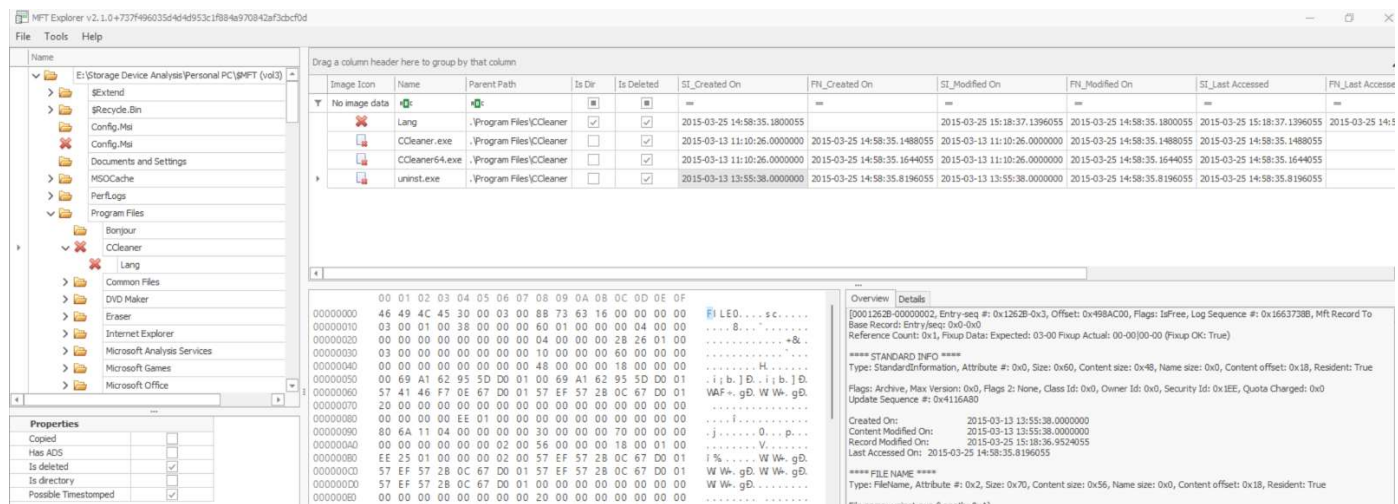
6.3. Web Search History

Inside his browser's web search history, we found an interesting topic that can interpret Mr Iaman's behaviour. On 23rd of March 2015, 07:07:58 UTC onward to 25th of March 2015, his search behaviour leaned towards "security concern" searches. Such as:

- E-mail investigation
- CD burning method in Windows
- Anti-forensic tools
- Eraser
- Ccleaner

This suggests the subject's behaviour is irregular.

We utilised MFT Explorer on the \$MFT, which contains valuable information about timestamps and deleted files on NTFS (Palmbach & Breiting, 2020). We were to confirm that this PC has downloaded CCleaner to help delete the files stored on it, including DVD Maker.



However, there is more to uncover from this deleted CCleaner.

Drag a column header here to group by that column										
SI_Created On	FN_Created On	SI_Modified On	FN_Modified On	SI_Last Accessed	FN_Last Accessed	SI_Record Changed	FN_Record Changed	Timestamped	Copied	
2015-03-25 14:58:35.1800055		2015-03-25 15:18:37.1396055	2015-03-25 14:58:35.1800055	2015-03-25 15:18:37.1396055	2015-03-25 14:58:35.1800055	2015-03-25 15:18:37.1396055	2015-03-25 14:58:35.1800055	☐	☐	
2015-03-13 11:10:26.0000000	2015-03-25 14:58:35.1488055	2015-03-13 11:10:26.0000000	2015-03-25 14:58:35.1488055	2015-03-25 14:58:35.1488055		2015-03-25 15:18:36.9368055	2015-03-25 14:58:35.1488055	☒	☐	
2015-03-13 11:10:26.0000000	2015-03-25 14:58:35.1644055	2015-03-13 11:10:26.0000000	2015-03-25 14:58:35.1644055	2015-03-25 14:58:35.1644055		2015-03-25 15:18:36.9368055	2015-03-25 14:58:35.1644055	☒	☐	
2015-03-13 13:55:38.0000000	2015-03-25 14:58:35.8196055	2015-03-13 13:55:38.0000000	2015-03-25 14:58:35.8196055	2015-03-25 14:58:35.8196055		2015-03-25 15:18:36.9524055	2015-03-25 14:58:35.8196055	☒	☐	

3 of the deleted files were marked as “Timestamped,” meaning the timestamps had been altered. This is a common anti-forensic technique. The same applies to Eraser.

Drag a column header here to group by that column

Image Icon	Name	Parent Path	Is Dir	Is Deleted	...	F...	SI_M...	...	SI_L...	FN...	SI...	FN_Record Changed	Timestamped	Copied
No image data														
	alibnet2.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	BevelLine.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	CommonLibrary.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	DragDropLib.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	Eraser Documentation.pdf	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	Eraser.exe	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...	2015-03-25 14:57:31.2718055	☒	☐
	Eraser.Manager.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	Eraser.Plugins.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐
	Eraser.Shell.dll	.Program Files\Eraser	☐	☐	...	2...	2015-...	...	201...	...	201...		☒	☐

```

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/PECmd

Command Line: -f E:\Storage Device Analysis\Personal PC\ERASER.EXE-CE61944A.pf

Warning: Administrator privileges not found!

Keywords: temp, tmp

Processing E:\Storage Device Analysis\Personal PC\ERASER.EXE-CE61944A.pf

Created on: 2025-11-04 06:12:45
Modified on: 2025-11-04 06:12:45
Last accessed on: 2025-11-04 06:15:03

Executable name: ERASER.EXE
Hash: CE61944A
File size (bytes): 153,292
Version: Windows Vista or Windows 7

Run count: 2
Last run: 2015-03-25 15:13:30

Volume information:

#0: Name: \DEVICE\HARDDISKVOLUME2 Serial: CA0C7A48 Created: 2015-03-25 11:08:36 Directories: 59 File references: 220
Directories referenced: 59

```

```

Select Administrator: Windows PowerShell

Processing E:\Storage Device Analysis\Personal PC\CCLEANER64.EXE-779BD542.pf

Created on: 2025-10-29 23:09:09
Modified on: 2025-10-29 23:09:09
Last accessed on: 2025-10-29 23:10:12

Executable name: CCLEANER64.EXE
Hash: 779BD542
File size (bytes): 29,954
Version: Windows Vista or Windows 7

Run count: 2
Last run: 2015-03-25 15:15:50

Volume information:

#0: Name: \DEVICE\HARDDISKVOLUME2 Serial: CA0C7A48 Created: 2015-03-25 11:08:36 Directories: 8 File references: 68
Directories referenced: 8

```


Using the PECmd, the Prefetch Cmd for parsing the Prefetch file in .pt file format. Good for extracting prefetch information of the running process (Neyaz & Shashidhar, 2022, pp. 207-208). We confirmed that this Ccleaner.exe and Eraser.exe were installed on 2015-03-25 11:08:36 UTC.

7. Limitation

Despite obtaining all the file images required to get started with forensic investigations, disk forensics proves useful if we were to analyse the whole storage media. Nonetheless, obtaining a live memory dump file would be helpful and provide substantial evidence to convict the suspect (Parekh & Jani, 2020).

A live memory dump holds valuable information, such as processes or any executable programs running at that moment. Therefore, significant information can be obtained by effectively analysing physical memory. The different types of information that can be extracted from memory include processes, dynamic link libraries (DLLs), process memory, image identification, kernel memory and objects, networking, registry, and malware.

8. Timeline



9. Conclusion

Initially, Mr. Spy Conspirator approached Mr. Iaman Informant via email communication using Outlook. They both exchanged a conversation that was vague; with enough evidence and artifacts gathered, we were able to piece together the meaning of their conversation.

With the motive of high payment in exchange for his company's data leakage, Mr. Iaman extracted "Seed files" into RM#2 (his second USB drive). Despite the DVD Maker program being found in his Program Files directory during \$MFT Explorer investigation, his RM#3 (CD-ROM) was not formatted, which still stores the same original files, indicating an attempt to format the CD (CD burn method) and failed due to the CD-ROM could not be written.

After file extraction onto the RM#2, Mr. Iaman executed both CCleaner and Eraser programs for deleting all of his evidence, including images, etc. More than 8000 worth of files. Later, the suspect attempted to uninstall CCleaner to destroy evidence.

Combined with his web browser search history for "Anti-forensic tools", he was able to perform "Timestomping", an anti-forensic technique that manipulates the actual timestamps of files. This action alone has proven his determination to hide and manipulate the evidence.

Notwithstanding the limitations of not obtaining a live memory dump, which would have eased the investigation and also provided substantial evidence in this case, we were able to gather enough evidence to convict the suspect of his actions.

These gathered evidence and computer artifacts have proven instrumental in establishing the culpability of Mr. Iaman Informant, as his actions contravened the organisation's security policies, resulting in the provision of third-party personnel with the company's confidential information without explicit authorisation.

References

- Boyanov, P. (2024). PRACTICAL APPLICATIONS OF HASH FUNCTIONS MD5, SHA-1, AND SHA-256 USING VARIOUS SOFTWARE TOOLS TO VERIFY THE INTEGRITY OF FILES: PRACTICAL APPLICATIONS OF HASH FUNCTIONS MD5, SHA-1, AND SHA-256 USING VARIOUS SOFTWARE TOOLS TO VERIFY THE INTEGRITY OF FILES. *Journal scientific and applied research*, 27(1), 120-137.
<https://doi.org/10.46687/jsar.v27i1.413>
- Jones, A., & Afrifa, I. (2020). An evaluation of data erasing tools. *Journal of Digital Forensics, Security and Law*, 15(1), 2. <https://doi.org/10.15394/jdfsl.2020.1615>
- Joshi, S., Kiashemshaki, K., & Roy, S. (2025, May). Comparing LNK File and Jump List Artifacts on Windows 11 with those on Windows 10. In *2025 IEEE International Conference on Electro Information Technology (eIT)* (pp. 378-382). IEEE.
<https://doi.org/10.1109/eIT64391.2025.11103693>
- Neyaz, A., & Shashidhar, N. (2022). Windows prefetch forensics. In *Breakthroughs in Digital Biometrics and Forensics* (pp. 191-210). Cham: Springer International Publishing.
<https://doi.org/10.1007/978-3-031-10706-1>
- Palmbach, D., & Breiting, F. (2020). Artifacts for detecting timestamp manipulation in NTFS on windows and their reliability. *Forensic Science International: Digital Investigation*, 32, 300920. <https://doi.org/10.1016/j.fsidi.2020.300920>
- Parekh, M., & Jani, S. (2020). Memory forensic: Acquisition and analysis of memory and its tools comparison. *International Journal of Engineering Technologies and Management Research*, 5(2), 90-95. <https://doi.org/10.5281/zenodo.1198968>
- Quick, D., & Choo, K. K. R. (2014). Google drive: Forensic analysis of data remnants. *Journal of Network and Computer Applications*, 40, 179-193.
<https://doi.org/10.1016/j.inca.2013.09.016>
- Yaacoub, J. P. A., Noura, H. N., Salman, O., & Chehab, A. (2021). Digital forensics vs. Anti-digital forensics: Techniques, limitations and recommendations. *arXiv preprint arXiv:2103.17028*. <https://doi.org/10.48550/arXiv.2103.17028>

Appendix A

Complete Email Conversation

This is gathered and pieced together into one streamlined email conversation.

23/3/2015

Spy: How are you doing?

Iaman: How are you doing?

Spy: Good Job. I need more detailed data about this business

Iaman: This is a sample.

Spy: Okay, I got it. I'll be in touch.

Spy: I confirmed it. But, I need a more data. Do your best

Iaman: Umm.... I need time to think.

Iaman: Use links below,

<https://drive.google.com/file/d/0Bz0ye6gXtiZaVl8yVU5mWHlGbWc/view?usp=sharing>

<https://drive.google.com/file/d/0Bz0ye6gXtiZaakx6d3R3c0JmM1U/view?usp=sharing>

Spy: I got it|

24/3/2015

Spy: This is the last request. I want to get the remaining data.

Iaman: Stop it! It is very hard to transfer all data over the internet!

Spy: No problem. U can directly deliver storage devices that stored it.

Iaman: This is the last time..

Spy: USB device may be easily detected. So, try another method

Iaman: I am trying.

Iaman: It's done. See you tomorrow.

Appendix B

Extracted Seed Files Hash Value Comparison

We have provided the “Extracted Seed Files” and the official hash value table for comparison. The label “X” shows in the hash table, indicating which hash value belongs to the “Extracted Seed Files”.

 Seed file extracted		11/14/2025 4:42 PM	File folder	
 Extracted Seed File Hash Value		11/4/2025 4:48 PM	Text Document	6 KB
 Hash Value Table		11/14/2025 4:53 PM	Microsoft Excel Work...	16 KB